actual empirical evidence proposed in favor of theories. This means a theory with a lot of empirical support will be scored as equal to a theory with almost no empirical support as long as they satisfy the same criteria. Similarly, CRIT does not take into account the amount of empirical evidence a theory's meeting of a criterion relies on. Theory A is scored as equal to theory B as long as they satisfy the same number of criteria, regardless of their respective sets of evidence. The second issue concerns *arbitration* between theories. Suppose two or more theories satisfy the same number of criteria, how do we decide between them? Given the limited number of criteria and the limited grading system on each criterion (e.g., table in Doerig et al., 2020, p. 48), the possibility of ties is high. *Arbitration* concerns not only how to decide between two or more theories that satisfy identical sets of criteria, but also how we should decide between two or more theories that satisfy the same number of criteria without their sets being identical. In other words, we need to know how to weigh satisfying criterion A against satisfying criterion B. CRIT is certainly useful for an overall classification of theories, but because it is not sensitive to divergent amounts of support, it is insufficient for any fine-grained comparison of theories. The third issue concerns the *flexibility* of the criteria. Now, Doerig and colleagues are explicit that the current set of criteria is not intended to be exhaustive (Doerig et al., 2020, p. 42) and will likely need expansion.[7] But how many — and which — criteria can we add? One might hope that the answer to this question is that any further criteria will be obvious, and we will come upon all — or most of — these over time (which in turn limits the maximum possible number of criteria as well). Observe, this answer may lead to a debate about what "obvious" entails, to whom it will be obvious, and who gets to decide these questions. This is the fourth issue: *arbitrariness*. For now, I will leave *arbitrariness* to the side since this issue will loom large throughout the text, and instead focus briefly on another upshot of *flexibility*; namely the question of how many criteria we will need to distinguish convincingly between theories (assuming we even can do this in a non-arbitrary way). Presently, any speculation on an exact number of further criteria would be premature. But given that the present set of criteria makes ties likely, it is likely to need expansion in the future. The next thing to note is that the set of criteria that are theory-neutral, obvious, overarching, and important is likely limited (however see, Rosenthal, 2021). This limitation would make any further criteria less central than the nine currently proposed. One reason for thinking this is that, if there were indeed further obvious and important criteria, Doerig and colleagues would have included them in their paper.[8] Be that as it may, it nevertheless is likely that a future expansion of CRIT will result in increasingly detailed criteria of less and less importance. One positive upshot of adding more criteria is that it seems CRIT may be able to deal with *arbitration* since ties will be less likely[9] as the number of criteria increases. However, this at the same time would undermine the main appeal of CRIT, i.e., identifying the *overarching principled* criteria a theory of consciousness should satisfy.

In the rest of this paper, I will present a third approach to assessing and comparing theories based on the notion of inference to the best explanation (IBE). Importantly, while I have identified shortcomings of both ARC and CRIT (*targeted theories, generalizability, robustness, cost* and *sensitivity, arbitration, flexibility, arbitrariness*, respectively), and will show that the approach proposed here does not have these shortcomings, I am not advocating that ARC and CRIT have no value, let alone should be given up. The approach here is intended to complement, rather than supplant, ARC and CRIT. There is room for these three approaches, not only to coexist, but to develop also a positive synergy. I will return to this in the concluding remarks.

3 Inference to the best explanation

In the previous section, I discussed two contemporary approaches to assessing and comparing theories. Previously, together with Peter Fazekas (Kirkeby-Hinrup and Fazekas, 2021), I have advocated a third approach based on the notion of inference to the best explanation. Looking at the publications over the last couple of decades, an IBE process seems tacit in much of the work concerned with the relation between empirical evidence and theories of consciousness. One of the most explicit invocations of IBE can be found in the work of Ned Block (2007, p. 486) when he says: "I have in mind […] the familiar default 'method' of inference to the best explanation, that is, the approach of looking for the framework that makes the most sense of all the data […]." Yet, to my knowledge, outside of my proposal with Fazekas, no one has endeavored to attempt inference to the best explanation in practice in ICS. One reason may be that classical versions of IBE are ill-suited for straightforward application in our situation.[10] This is because we cannot compare theories on their explanatory powers, because there is no consensus on a common explanandum. To elaborate, competing theories do not necessarily have identical explanatory targets (Sattin et al., 2021; Signorelli et al., 2021; Yaron et al., 2021), yet are taken to be mutually exclusive for the reason that they all target the same phenomenon (and they share the assumption that there is only one phenomenon). In the vocabulary of Chalmers (2002), theories differ in their 'intension' of the explanandum (the meaning of the word 'consciousness'), but coincide on its 'extension' (the thing in the world picked out by the word 'consciousness'). In a way, when deploying empirical evidence in assessing and comparing theories of consciousness, we are hoping to resolve disagreements on the *intension* through investigations of the *extension*. The upshot is that we cannot adopt explanatory power as our metric for comparison, since explanatory power depends on the 'intension' of the explanandum, which means we would be comparing apples and oranges. Therefore, we must perform IBE on the basis of some other metric than explanatory power. One way to approach this is by collating the respective sets of proposed empirical support of the competing theories, to determine if our observations about the *extension* (empirical evidence) conform to a proposed *intension* (a theory), and how well.

The notion of IBE (sometimes understood as co-extensive with the notion of abduction) is a classic topic in the philosophy of science (Burks, 1946; Harman, 1965; Peirce and Hartshorne, 1974;

7 There are already candidates for further criteria (Overgaard and myself have proposed one Kirkeby-Hinrup and Overgaard, 2023).

8 This is tenuous of course, given that there may be a range of other reasons elements were not included in a paper.

9 Assuming it would not make sense to add a criterion that every theory satisfies.

10 Another plausible reason is the lack of datasets necessary to carry out IBE, see Section 4.

Minnameier, 2004, 2010; Campos, 2011; Douven, 2021). But, for the reason just given, classical notions are not straightforwardly applicable in our case. Therefore, some clarification is necessary with respect to the way IBE is conceived of here. Firstly, the 'explanations' we need to infer to are theories of consciousness (what Block called "frameworks" in the quote above. In Chalmers' vocabulary, the different proposed *intensions* of consciousness). Secondly, the assessment and comparison are not based on *explanatory* considerations, *per se*. The metric for assessment — and what is being compared — is not a theory's explanatory power in relation to its targeted explanandum (its *intension*). As just noted, comparing explanatory power in relation to *intension* of the explanandum is problematic because there is no agreement on what a good explanation would entail, because there is no agreement on the exact characteristics of the phenomenon (see, e.g., debate in Rosenthal, 2011; Weisberg, 2011; Block, 2011a,c). To avoid this, the IBE approach could consist in assessing and comparing (i.e., inferring on the bases of) the explanatory power of theories in the empirical domain. In other words, the metric of comparison in this proposal is the ability to explain and predict empirical data.

There are many ways to develop an IBE process. Fazekas and I (Kirkeby-Hinrup and Fazekas, 2021) proposed a four-step process relying on the fact that the first step (assimilation, i.e., data collection) was already far along,[11] argued the importance — and demonstrated the feasibility — of the second and third step: compilation and validation (respectively concerned with compiling the proposed evidence for each theory, and validating claims of empirical support on a case-by-case basis). To elaborate, in addition to demonstrating that the second and third steps are feasible by showing what they look like in practice, the main point of the paper was that if we want to decide between theories on the bases of their respective empirical support, we had better know what their respective empirical support is,[12] and whether any given piece of empirical support claimed by a theory, in fact supports the theory. The step of the IBE process that is developed below is the one we did not treat in that paper, namely the actual comparison of theories,[13] the fourth and final step. Since the proposal here depends on quantifying the competing sets of proposed evidence, I will call this approach *Quantification to the Best Explanation* (QBE).

3.1 An intuition about weights of evidence

An initial desideratum is that QBE should avoid the identified shortcomings of ARC and CRIT (*targeted theories, generalizability, robustness, cost*, *sensitivity, arbitration, flexibility,* and *arbitrariness*) discussed in the previous section. While QBE avoids many of these easily, two warrant consideration here,[14] namely: *sensitivity* and *arbitration*. These two shortcomings appear to threaten QBE and CRIT equally. To clarify, the sets of empirical evidence proposed for the extant theories of consciousness are *prima facie* incommensurable. One source of the incommensurability is that the sets of empirical support for each of the theories — while in many cases partially overlapping — do not contain exactly the same elements. Thus, the *arbitration* issue reappears on IBE, because we now need a way to weigh the non-overlapping elements against each other. To illustrate, it is unclear whether theory A being supported by the change blindness phenomenon is more "valuable" (as it were) than theory B's support from the split-brain phenomenon. Yet, many share the intuition that *some* instances of empirical evidence should weigh heavier than others. This raises at least two questions: First: what is the driver of this intuition? And second: which instances? Let us focus here on the first question (the second question will be addressed in subsequent sections). Leaving open whether there are others, here are at least two possible candidate drivers of the intuition that some evidence is more "valuable" (should weigh heavier in IBE) than other (Figure 1).

The first candidate as a driver is that the 'closer' (applicable) a piece of evidence (a phenomenon) is to the normal human condition (i.e., consciousness *as such,* or consciousness in neurotypical adults) the higher weight it should be ascribed in a comparison process. Call this the *closeness* driver. Closeness could be understood as *physical/functional* closeness, suggesting that studies with human subjects are more "valuable" than animal studies or computational models. Another example of *physical/functional* closeness could be the intuition that studies on neurotypical brains are preferable to studying very rare cases of brain trauma or cognitive dysfunction. Another way to understand *closeness* could be as *distribution*, which can be subdivided into inter-individually and temporally, where the former tracks the number of individuals to which the phenomenon applies and the latter tracks *how often* an individual or group of individuals instantiate the phenomenon. Accordingly, phenomena with high

11 We suggested the first stage — in which proponents of theories collected empirical support — had already been ongoing for a couple of decades, and therefore there was no need to demonstrate its feasibility.

12 Given the complexity of debates between competing theories of consciousness, it is not sufficient to select reports in which empirical evidence is proposed explicitly in support of a theory. In fact, significant (and increasing) parts of the academic exchange between proponents of competing theories take place in so-called proxy debates. Proxy debates, as the name indicates, are not directly about the theories. Instead, they are about specific empirical phenomena or aspects of consciousness, on which the positions taken in the debate are (sometimes tacit) extensions of central views of particular theories. Therefore, when proxy debates deploy empirical evidence, this should be seen as part of the empirical support for a theory. Over the last couple of decades the occurrence of such proxy debates has been relatively steady. Examples include the debates about unconscious perception (Brogaard, 2011; Block, 2016; Peters et al., 2017), non-conceptual content (Brinck, 1999; Jacobson and Putnam, 2016), whether perception is rich or sparse (Kouider et al., 2010; Block, 2011b, 2014b; Knotts et al., 2019), and perceptual precision (Block, 2014a; Prettyman, 2019). A prominent ongoing proxy debate concerns the localization of the neural correlates of consciousness (usually called either the "front vs. back" debate or the "early vs. late" debate). For a small sample of this debate see, e.g., (Lamme, 2003, 2004; Bor and Seth, 2012; Meuwese et al., 2013; Frässle et al., 2014; Kozuch, 2014; Boly et al., 2017; Odegaard et al., 2017; Michel and Morales, 2020).

13 While the methodology proposed here caters to the fourth step and thereby complete the account offered by Fazekas and me, nothing in the below hinges on acceptance of our claims in that paper. Readers uninclined to this view of an IBE process may nevertheless find use for a methodology for quantifying empirical evidence and comparing sets of evidence.

14 The rest will be addressed in the concluding remarks.

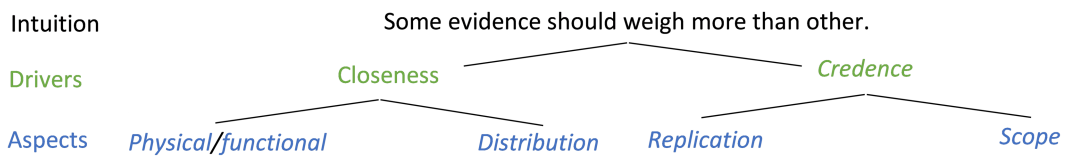

FIGURE 1
The intuition and two drivers.

distribution (inter-individually, temporally, or both) are 'closer' to the explanatory target ('neurotypical adult consciousness' because *many* experience the phenomenon *often*) and consequently should be given higher weight in the IBE process.

The second candidate driver of the intuition (that some evidence should weigh heavier than other) concerns our *credence* in the evidence in question. Call this the *credence* driver. According to *credence*, the extent of our knowledge of a given phenomenon seems to matter for the weight it should be ascribed in the IBE process. *Credence*, furthermore, can be subdivided at least into *replication* and *scope*. *Replication* concerns the robustness of our ways of knowing about the phenomenon, i.e., the total amount of studies conducted on it, the amount of replication studies, and the existence of well-established paradigms to investigate it. *Scope* concerns the number of angles we (could) have approached the phenomenon from, i.e., the range of empirical techniques (that can be) applied to it. To elaborate; *replication* considers that phenomena, that have been the subject of thousands of studies and on the basic features of which (independent of any specific theory of consciousness) there is a general consensus, are more valuable than phenomena that have only been recorded very few times and the interpretations of which are widely contentious *outside* of consciousness studies. *Scope*, on the other hand, concerns the number of empirical techniques that have been — or could be — used to investigate the phenomenon. According to *scope*, phenomena that have been measured in many ways (e.g., fMRI, EEG, PET, MEG, ECoG, fNIRS, eye blinks, saccades, eye-tracking, D′, Meta D′, reaction time, introspective report, perceptual awareness scale, to name a few) are more valuable than phenomena that only have been — or only can be — measured using a single or few techniques (e.g., phenomena relying solely on introspective report). Thus — overall on *credence* — the intuition would be that phenomena with either high replication, broad scope, or both should be given higher weight in QBE.[15]

4 Bayesian inference to the best explanation

If we follow the intuition that some evidence should weigh heavier than other, a second shortcoming of the existing approaches that also is a challenge for QBE is *arbitrariness*. In this context, *arbitrariness* concerns *who* gets to assign the weights to the pieces of empirical evidence, and whether this can be done in a theory-neutral and non-contentious way. The "who" matters because, if the assignment of weights in QBE depends arbitrarily on the person performing the comparison, the objectivity of the process is compromised which, for obvious reasons, would be a bad thing. Consequently, a second desideratum for QBE is that it can deliver an objective way (one that does not depend on arbitrary choices of the person performing the comparison) to ascribe weight to proposed empirical support.

As an starting point, Fazekas and I (Kirkeby-Hinrup and Fazekas, 2021) suggested that— in order to stay neutral between theories when evaluating evidence — it is preferable to evaluate each piece of evidence in light of the conceptual framework of the theory it is applied to. This is because using any other conceptual framework ('intension' of consciousness) risks begging the question against the theory, i.e., presuming a viewpoint and thereby giving up on objectivity.

From this starting point, since we are aiming to quantify empirical support, we need a way to get numbers, via the sets of empirical evidence proposed in favor of extant theories. The conversion to numbers is made difficult by the way empirical evidence is proposed in ICS, *viz* what we are trying to quantify is really arguments to the effect that some piece of empirical evidence is predicted by — or can be explained in light of — a given theory. Such arguments, in turn, depend on the conceptual framework of the theory, and the mapping of an interpretation of some empirical data to this framework (see Kirkeby-Hinrup and Fazekas, 2021 for details).[16] This kind of

15 Observe, I do not claim that *closeness* and *credence* are the only possible sources for an intuition that some instances of empirical support should be given more weight than others when comparing theories of consciousness. It is entirely possible that there are other sources for this intuition. Nevertheless, since this intuition appears to be rooted at least in a combination of our areas of interest (*closeness*) and scientific principles (*credence*), I will assume we can agree on one or more of the drivers *physical/functional closeness, distribution, replication, and scope*. Importantly, my proposal does not depend on whether the reader subscribes to this intuition (or all of its possible drivers). QBE can be deployed independently of this intuition (for instance as a feasible theory neutral way to assess and compare theories of consciousness, that does not have the shortcomings of CRIT and ARC).

16 Relevant questions in this regard include whether the same concepts are applied to — or operationalized in (Fink, 2016) — the interpretation in a uniform way. Whether there are equivocations or vagueness in the application of terms from the conceptual framework. It is crucial correctly to identify the theoretical claim defended, since this has implications for the way the argument is evaluated. Plainly, to assess an argument it is imperative to identify what it is an argument *for*. It matters for the assessment whether an argument is about overflow or recurrent processing (even if both pertain to RPT). To understand an argument in this context is to investigate how the proposed interpretations of the empirical evidence map onto the theoretical principle or concept, and illuminating (by extrapolation, if necessary) the argument connecting the